# Standardized and No-Intercept Models

Applied Linear Models I • Lecture 8 of 16

# Today (then the midterm)

- **Standardized** (beta) coefficients: per-SD effects.
- The identity $\beta_j^* = \beta_j (s_{x_j}/s_y)$; centering kills the intercept.
- Comparing predictor **importance** across units.
- **No-intercept** regression: when valid, what changes if misused.

The midterm covers Lectures 1–8.

# Standardized regression

Replace each variable by its $z$-score $(v - \bar{v})/s_v$ and refit.

**Interpretation**

$\beta_j^*$ is the change in $y$, **in SDs of $y$**, per one-SD increase in $x_j$ (others fixed) – the *same unit* for every predictor.

Scaling identity (verified to machine precision):

$$\beta_j^* = \beta_j \frac{s_{x_j}}{s_y}.$$

# Centering kills the intercept

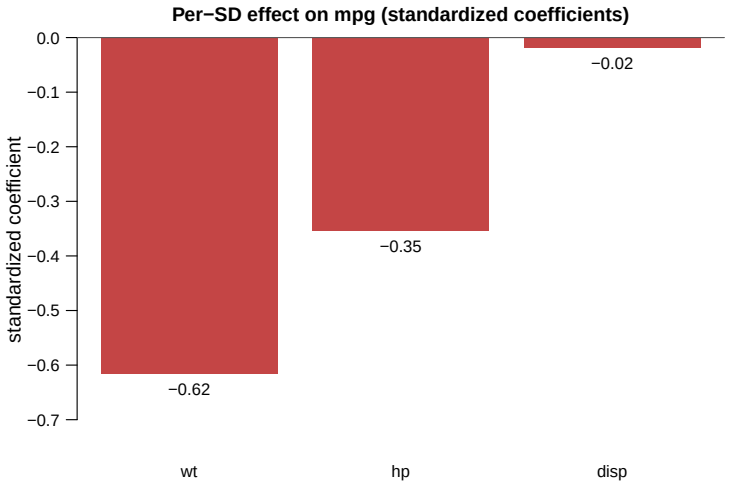
Both $y$ and the $x_j$ are centered $\Rightarrow$ the fit passes through the origin:

$$\hat{\beta}_0^* = 3.6 \times 10^{-17} \approx 0.$$

- Standardized regressions are often written without an intercept.
- Centering improves the conditioning of $X^\top X$ for the slopes (relevant to Lecture 10) – but does **not** cure collinearity.

# Comparing predictor importance

Raw coefficients in different units are **not comparable**. Standardized ones are:



# Caveats on "importance"

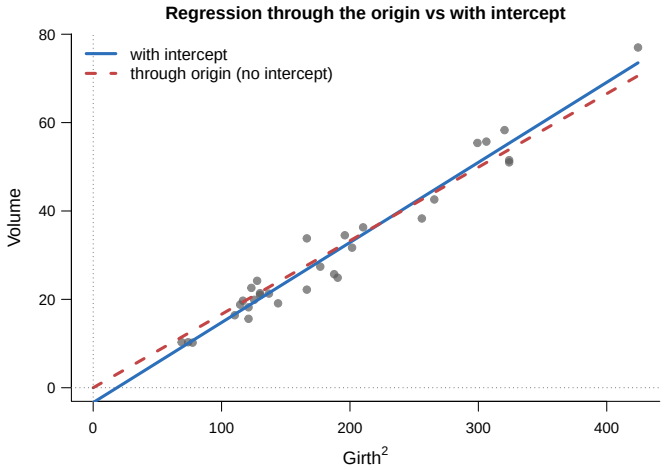
Standardized coefficients measure importance *within this sample and model*:

- sensitive to the predictor's **sampled variance** (restricted range $\Rightarrow$ smaller $\beta^*$);
- distorted by **collinearity**;
- depend on which other predictors are included.

A useful comparison device – not a definitive causal ranking.

# Regression through the origin

Drop the intercept **only** when theory forces $\mathbb{E}[y] = 0$ at $x = 0$.



# What changes without an intercept

1. **Residuals need not sum to zero**: $1 \notin \mathcal{C}(X)$, so $1^\top e \neq 0$ (here $\sum e_i \approx -19$).
2. **$R^2$ is not comparable**: a different baseline, often misleadingly high.
3. **Bias** if the true intercept is nonzero.

**Honest tension**

Theory says drop it, but the data give a marginally significant intercept ($p \approx 0.025$). **Default: keep the intercept.**

# Summary & midterm

- $\beta_j^* = \beta_j(s_{x_j}/s_y)$: per-SD effects, common scale; intercept $\to 0$.
- Compare importance across units – with caveats.
- No-intercept: only when theory demands; changes residuals, $R^2$; can bias.

**Midterm**

Covers Lectures 1–8. Next: Lecture 9, polynomial and interaction models.